

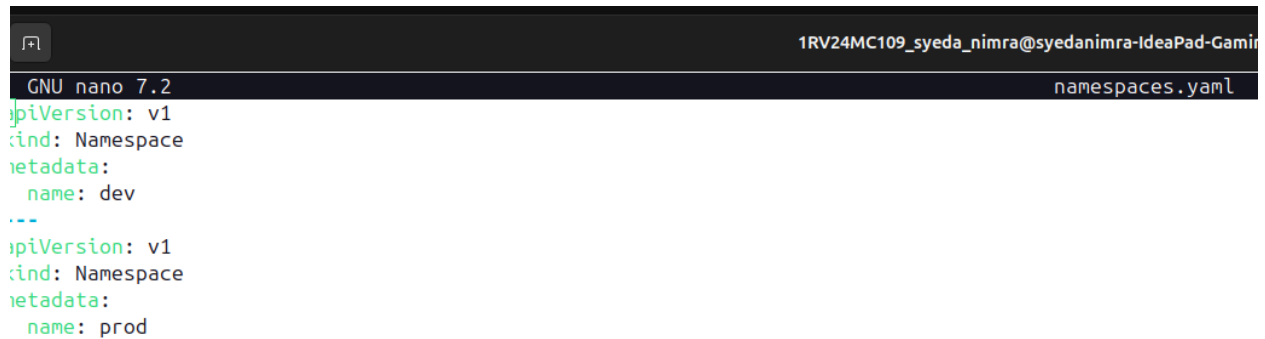
## Program-9: Steps to Execute

### Step 1: Create Project Directory and start minikube

```
mkdir Program-9  
cd Program-9
```

### Step 2: Create namespaces.yaml

```
nano namespaces.yaml
```



```
GNU nano 7.2 namespaces.yaml  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: dev  
---  
apiVersion: v1  
kind: Namespace  
metadata:  
  name: prod
```

### Step 3: Create pods.yaml

```
nano pods.yaml
```

```
1RV24MC109_syeda_nimra@syedanimra-IdeaPad-Gaming-3-15IMH05:
GNU nano 7.2 pods.yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: dev
  labels:
    app: frontend
    env: dev
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
--
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: dev
  labels:
    app: backend
    env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
--
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
```

[ Read 60 lines ]

|         |              |             |          |            |               |          |    |
|---------|--------------|-------------|----------|------------|---------------|----------|----|
| ^G Help | ^O Write Out | ^W Where Is | ^K Cut   | ^T Execute | ^C Location   | M-U Undo | M- |
| ^X Exit | ^R Read File | ^_ Replace  | ^U Paste | ^J Justify | ^_ Go To Line | M-E Redo | M. |

## Step 4: Apply Namespace File and Pods File

kubectl apply -f namespaces.yaml

kubectl apply -f pods.yaml

## Step 5: Verify Pods by Namespace

kubectl get pods -n dev

kubectl get pods -n prod

```
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl apply -f namespaces.yaml
```

```
namespace/dev unchanged
```

```
namespace/prod unchanged
```

```
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl apply -f pods.yaml
```

```
pod/frontend-pod unchanged
```

```
pod/backend-pod unchanged
```

```
pod/frontend-pod unchanged
```

```
pod/backend-pod unchanged
```

```
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl get pods -n dev
```

```
kubectl get pods -n prod
```

| NAME         | READY | STATUS  | RESTARTS     | AGE |
|--------------|-------|---------|--------------|-----|
| backend-pod  | 1/1   | Running | 5 (7h9m ago) | 19d |
| frontend-pod | 1/1   | Running | 5 (7h9m ago) | 19d |

| NAME         | READY | STATUS  | RESTARTS     | AGE |
|--------------|-------|---------|--------------|-----|
| backend-pod  | 1/1   | Running | 5 (7h9m ago) | 19d |
| frontend-pod | 1/1   | Running | 5 (7h9m ago) | 19d |

```
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl get pods
```

| NAME                              | READY | STATUS  | RESTARTS     | AGE   |
|-----------------------------------|-------|---------|--------------|-------|
| demo-pod                          | 1/1   | Running | 1 (7h9m ago) | 7h53m |
| labeled-pod                       | 1/1   | Running | 1 (7h9m ago) | 7h30m |
| nginx-deployment-6f9664446b-sckkm | 1/1   | Running | 0            | 43m   |
| nginx-deployment-6f9664446b-tj72l | 1/1   | Running | 0            | 43m   |
| simple-pod                        | 1/1   | Running | 1 (7h9m ago) | 7h37m |

## Step 6: Filter Pods using Labels

List frontend pods in dev: `kubectl get pods -n dev -l app=frontend`

List backend pods in prod: `kubectl get pods -n prod -l app=backend`

```
simple-pod 1/1 Running 1 (7h9m ago) 7h37m
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl get pods -n dev -l app=frontend

kubectl get pods -n prod -l app=backend
NAME READY STATUS RESTARTS AGE
frontend-pod 1/1 Running 5 (7h10m ago) 19d
NAME READY STATUS RESTARTS AGE
backend-pod 1/1 Running 5 (7h10m ago) 19d
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$ kubectl get pods -n prod -l app=backend
NAME READY STATUS RESTARTS AGE
backend-pod 1/1 Running 5 (7h12m ago) 19d
RV24MC109_syeda_nimra@syedaninra-IdeaPad-Gaming-3-15IMH05:~/prog9$
```